EDGE Intelligence LAB Assignment (26-03-2026)

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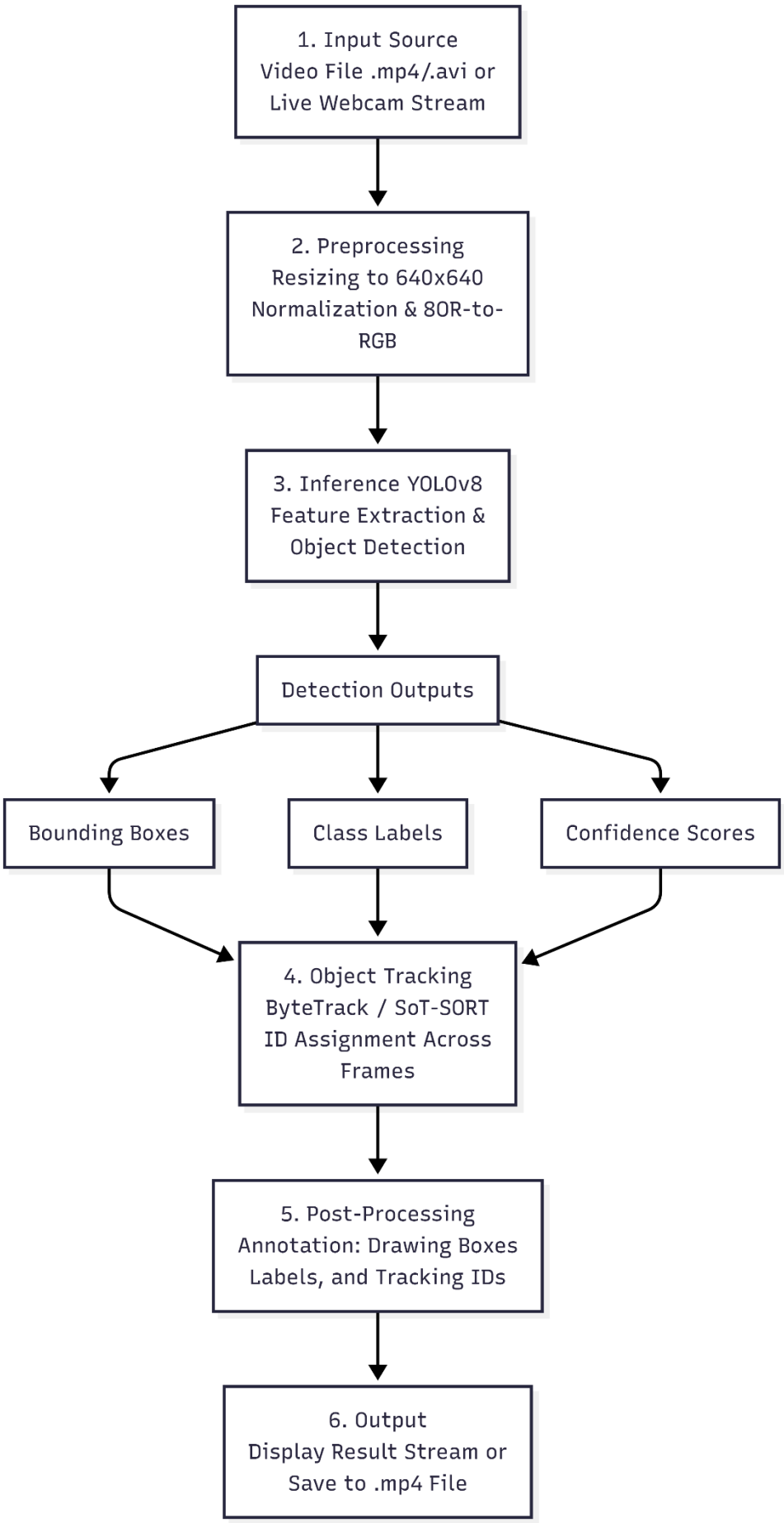
25MML0029

Pipeline for Object Detection in Videos Using Yolov8 and Coding

Pipeline:

Done Using Mermaid


1. Input Source
Video File .mp4/.avi or
Live Webcam Stream
2. Preprocessing
Resizing to 640x640
Normalization & BOR-to-
RGB
3. Inference YOLOv8
Feature Extraction &
Object Detection
Detection Outputs
Bounding Boxes
Class Labels
Confidence Scores
4. Object Tracking
ByteTrack / SoT-SORT
ID Assignment Across
Frames
5. Post-Processing
Annotation: Drawing Boxes
Labels, and Tracking IDs
6. Output
Display Result Stream or
Save to .mp4 File

Code :

[2]
5s

```
!pip install ultralytics
```

```
Collecting ultralytics
  Downloading ultralytics-8.4.30-py3-none-any.whl.metadata (39 kB)
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.2)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (3.10.0)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (4.13.0.92)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.16.3)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.10.0+cpu)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.25.0+cpu)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.35.2)
Collecting ultralytics-thop>=2.0.18 (from ultralytics)
  Downloading ultralytics_thop-2.0.18-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (26.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (2.9.0.post0)
Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.12/dist-packages (from polars>=0.20.0->ultralytics) (1.35.2)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2026.2.25)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (4.15.0)
```

[3]
54s

```
import cv2
from ultralytics import YOLO
from google.colab.patches import cv2_imshow

# 1. Load the model
model = YOLO("yolov8n.pt")

# 2. Open your video
video_path = "/content/Traffic.55.33 PM.mp4"
cap = cv2.VideoCapture(video_path)

# Get properties to ensure the output matches the input exactly
w = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
h = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# 3. Setup VideoWriter
# 'mp4v' or 'avc1' are standard for .mp4 files
fourcc = cv2.VideoWriter_fourcc(*'mp4v')
out = cv2.VideoWriter("output_video.mp4", fourcc, fps, (w, h))

while cap.isOpened():
    success, frame = cap.read()
    if not success:
        break

    # Run tracking (stream=True is more memory efficient for videos)
    results = model.track(frame, persist=True, stream=True)

for r in results:
    # Plot boxes and IDs on the frame
    annotated_frame = r.plot()

    # Write the processed frame to the video file
    out.write(annotated_frame)

cap.release()
out.release()
cv2.destroyAllWindows()
```

```
Creating new Ultralytics Settings v0.0.6 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralytics.com/quickstart/#ultralyt
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt to 'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.2MB 73.4MB/s 0.1s
requirements: Ultralytics requirement ['lap>=0.5.12'] not found, attempting AutoUpdate...
Using Python 3.12.13 environment at: /usr
Resolved 2 packages in 226ms
Prepared 1 package in 101ms
Installed 1 package in 1ms
 + lap==0.5.13

requirements: AutoUpdate success ✅ 0.8s
WARNING ⚠️ requirements: Restart runtime or rerun command for updates to take effect


0: 640x384 1 person, 10 cars, 9 buss, 284.1ms
Speed: 2.6ms preprocess, 284.1ms inference, 43.5ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 1 person, 9 cars, 9 buss, 101.7ms
Speed: 1.5ms preprocess, 101.7ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 1 person, 9 cars, 9 buss, 92.0ms
Speed: 1.2ms preprocess, 92.0ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 1 person, 9 cars, 9 buss, 107.2ms
Speed: 1.7ms preprocess, 107.2ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 1 person, 9 cars, 9 buss, 97.0ms
Speed: 1.4ms preprocess, 97.0ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 1 person, 9 cars, 9 buss, 94.7ms
Speed: 3.2ms preprocess, 94.7ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 384)
```

```
0: 640x384 1 person, 9 cars, 9 buss, 95.6ms
Speed: 1.4ms preprocess, 95.6ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 1 person, 9 cars, 9 buss, 1 truck, 93.7ms
Speed: 1.2ms preprocess, 93.7ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 1 person, 9 cars, 9 buss, 1 truck, 99.0ms
Speed: 0.9ms preprocess, 99.0ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 1 person, 9 cars, 9 buss, 1 truck, 93.6ms
Speed: 1.9ms preprocess, 93.6ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 8 cars, 9 buss, 1 truck, 97.5ms
Speed: 1.0ms preprocess, 97.5ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 8 cars, 9 buss, 1 truck, 104.5ms
Speed: 2.6ms preprocess, 104.5ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 7 cars, 9 buss, 1 truck, 94.2ms
Speed: 1.7ms preprocess, 94.2ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 7 cars, 9 buss, 1 truck, 93.2ms
Speed: 1.9ms preprocess, 93.2ms inference, 0.8ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 7 cars, 8 buss, 1 truck, 94.6ms
Speed: 1.6ms preprocess, 94.6ms inference, 0.9ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 8 cars, 8 buss, 1 truck, 97.8ms
Speed: 1.4ms preprocess, 97.8ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 384)

0: 640x384 9 cars, 7 buss, 1 truck, 99.9ms
Speed: 2.2ms preprocess, 99.9ms inference, 0.7ms postprocess per image at shape (1, 3, 640, 384)
```

Input Video Frame:

Output Video frame: